Tobias Mwangi-Hartley

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PROFESSIONAL SUMMARY

Data Scientist with experience developing applied machine-learning workflows for forecasting and customer retention, from feature engineering and model evaluation through scheduled notebook jobs and batch scoring. Builds scheduled-notebook forecasting and batch-scoring workflows with Python, BigQuery and Vertex AI Workbench, plus a Dockerised Streamlit experimentation application. Experience includes feature engineering, model evaluation, pytest testing and operational delivery to Looker; currently transitioning from applied Data Science toward production ML engineering. MSc in Statistics with a dissertation on Bayesian hierarchical models.

TECHNICAL SKILLS

Programming and analysis: Python, SQL, R, pandas, scikit-learn
Machine learning: LightGBM, XGBoost, feature engineering, forecasting, churn modelling, SHAP
Data, cloud and workflow tools: BigQuery, Vertex AI Workbench, dbt, dbt Cloud scheduler, Looker
Software, testing and experimentation: Python, pytest, Docker, Streamlit, statsmodels, A/B testing

PROFESSIONAL EXPERIENCE

Data Scientist | Thistlebox *2022–Present*

- Develop forecasting, experimentation and churn-modelling solutions for product and operations teams.
- Built a weekly recipe-demand forecasting workflow using LightGBM and calendar, price and menu-position features; ran it as a scheduled Vertex AI Workbench notebook, wrote six-week forecasts to BigQuery, and made results available to operations through Looker.
- Improved forecast WAPE from 23% to 14%; estimated food-waste reduction was 9%.
- Developed an XGBoost churn-propensity model with Python, SHAP and feature engineering; scored customers weekly in BigQuery using dbt-supported workflows and targeted retention offers to the top decile. A 10% holdout evaluation reported a 2.1 percentage-point reduction in churn in the treated group.
- Built a Streamlit application for planning and analysing sequential A/B tests, used by approximately 25 people in product and growth. Authored its Dockerfile; the data platform team handles deployment to Cloud Run.
- Since 2025, writing pytest unit tests for forecasting feature-engineering code; current coverage is approximately 60%.
- Delivered internal “Stats for PMs” training three times.

Junior Data Analyst | Firth Energy Retail *2020–2022*

- Segmented customers and analysed tariffs using R and SQL.
- Automated monthly regulator reports with R Markdown, saving the team approximately two days each month.

SELECTED PROJECT

Image Classification Course Project | Personal learning project *2025*

- Fine-tuned a small CNN on a public food-images dataset, achieving approximately 80% accuracy as part of an online deep-learning course.
- Tools: PyTorch, torchvision, Google Colab.

EDUCATION

MSc Statistics
Dissertation: Bayesian hierarchical models